

NCP-AIN AI Fabric Pro Mega Lab - Topology 2 Clear Poster

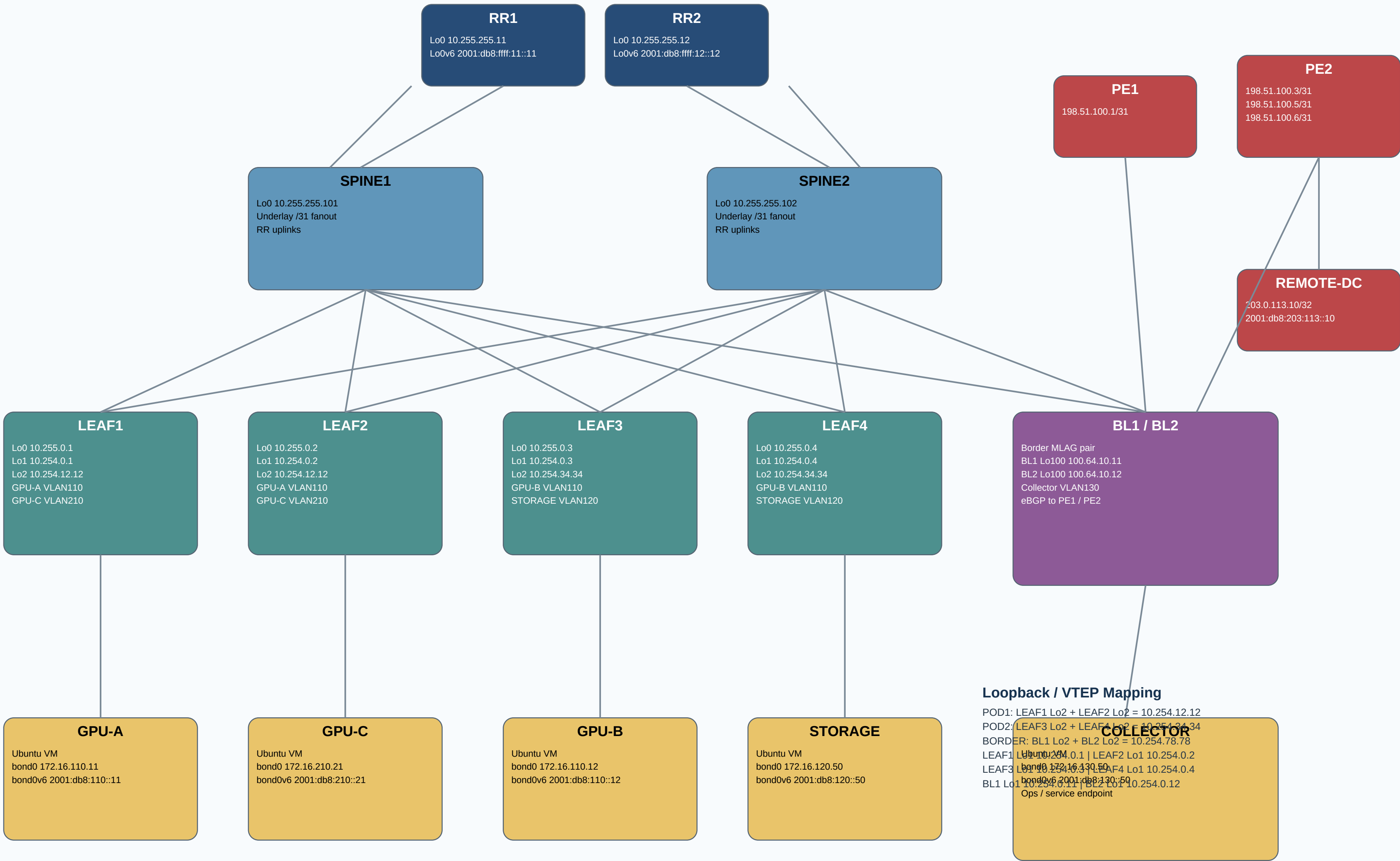
Clear PNG-friendly view with separate mapping tables for ports, /31 links, VLAN/VNI, and IPv6

Underlay /31 Mapping

LEAF1 E1 10.0.0.0/31 <-> SPINE1 E1 10.0.0.1/31
LEAF1 E2 10.0.0.12/31 <-> SPINE2 E1 10.0.0.13/31
LEAF2 E1 10.0.0.2/31 <-> SPINE1 E2 10.0.0.3/31
LEAF2 E2 10.0.0.14/31 <-> SPINE2 E2 10.0.0.15/31
LEAF3 E1 10.0.0.4/31 <-> SPINE1 E3 10.0.0.5/31
LEAF3 E2 10.0.0.16/31 <-> SPINE2 E3 10.0.0.17/31
LEAF4 E1 10.0.0.6/31 <-> SPINE1 E4 10.0.0.7/31
LEAF4 E2 10.0.0.18/31 <-> SPINE2 E4 10.0.0.19/31
BL1 E1 10.0.0.8/31 <-> SPINE1 E5 10.0.0.9/31
BL1 E2 10.0.0.20/31 <-> SPINE2 E5 10.0.0.21/31
BL2 E1 10.0.0.10/31 <-> SPINE1 E6 10.0.0.11/31
BL2 E2 10.0.0.22/31 <-> SPINE2 E6 10.0.0.23/31
RR1 E1 10.0.100.1/31 <-> SPINE1 E10 10.0.100.0/31
RR1 E2 10.0.100.3/31 <-> SPINE2 E10 10.0.100.2/31
RR2 E1 10.0.100.5/31 <-> SPINE1 E11 10.0.100.4/31
RR2 E2 10.0.100.7/31 <-> SPINE2 E11 10.0.100.6/31

Host Attachment Mapping

GPU-A eth0 -> LEAF1 E20 | eth1 -> LEAF2 E20 | Po10 | VLAN110
GPU-C eth0 -> LEAF1 E21 | eth1 -> LEAF2 E21 | Po11 | VLAN210
GPU-B eth0 -> LEAF3 E20 | eth1 -> LEAF4 E20 | Po20 | VLAN110
STORAGE eth0 -> LEAF3 E21 | eth1 -> LEAF4 E21 | Po21 | VLAN120
COLLECTOR eth0 -> BL1 E20 | eth1 -> BL2 E20 | Po30 | VLAN130
POD1 peer-link: LEAF1/2 E10-E11 -> Port-Channel100
POD2 peer-link: LEAF3/4 E10-E11 -> Port-Channel100
BORDER peer-link: BL1/2 E10-E11 -> Port-Channel100



Loopback / VTEP Mapping

POD1: LEAF1 Lo2 + LEAF2 Lo2 = 10.254.12.12
POD2: LEAF3 Lo2 + LEAF4 Lo2 = 10.254.34.34
BORDER: BL1 Lo2 + BL2 Lo2 = 10.254.78.78
LEAF1 Ubuntu VM 0.1 | LEAF2 Lo1 10.254.0.2
LEAF3 Lo1 10.254.0.3 | LEAF4 Lo1 10.254.0.4
BL1 Lo1 10.254.0.1 | BL2 Lo1 10.254.0.12
Ops / service endpoint

External Edge Mapping

BL1 E30 198.51.100.0/31 -> PE1 E1 198.51.100.1/31
BL2 E30 198.51.100.2/31 -> PE2 E1 198.51.100.3/31
PE1 E2 198.51.100.4/31 -> PE2 E2 198.51.100.5/31
PE2 E3 198.51.100.6/31 -> REMOTE E1 198.51.100.7/31
REMOTE-DC Lo100 IPv4 203.0.113.10/32
REMOTE-DC Lo100 IPv6 2001:db8:203:113::10/128

Dual-Stack VLAN / VNI / Gateway Mapping

VLAN110 -> VNI10110 -> VRF AI_FABRIC
IPv4 subnet 172.16.110.0/24 | GW 172.16.110.1
IPv6 subnet 2001:db8:110::/64 | GW 2001:db8:110::1
VLAN120 -> VNI10120 -> VRF AI_FABRIC
IPv4 subnet 172.16.120.0/24 | GW 172.16.120.1
IPv6 subnet 2001:db8:120::/64 | GW 2001:db8:120::1
VLAN130 -> VNI10130 -> VRF AI_SERVICES
IPv4 subnet 172.16.130.0/24 | GW 172.16.130.1
IPv6 subnet 2001:db8:130::/64 | GW 2001:db8:130::1
VLAN210 -> VNI10210 -> VRF TENANT_B